MERIDIAN JUNIOR COLLEGE
2009 JC2 PRELIMINARY EXAMINATION

MATHEMATICS **9740/02**
Higher 2
Paper 2

Thursday 17 September 2009
3 hours

Additional materials: Writing paper
List of Formulae (MF 15)

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This document consists of **7** printed pages.

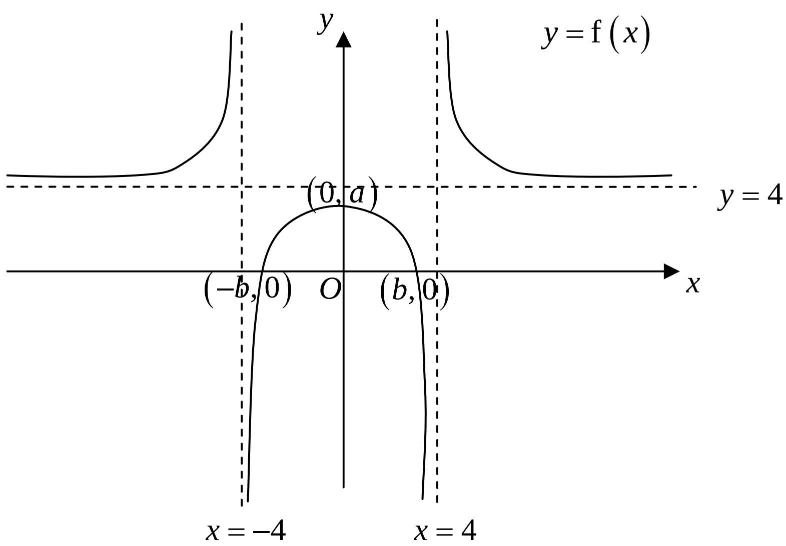
Section A: Pure Mathematics [40 marks]

1 **(i)** Differentiate $e^{\cos x}$ with respect to x. [1]

(ii) Find $\int e^{\cos x} \sin 2x \, dx$. [3]

2 **(a)** Sketch the graph of $y = \dfrac{2x+11}{(x+3)^2}$, showing clearly the asymptotes, axial intercepts and coordinates of turning points, if any. [4]

(b)



The diagram above shows the graph of $y = f(x)$. The curve has x-intercepts at $(-b, 0)$ and $(b, 0)$ and a maximum point at $(0, a)$ where $0 < a < 4$ and $1 < b < 4$. On separate diagrams, sketch the graphs of

(i) $y = f(2x-1)$, [3]

(ii) $y^2 = f(x)$. [2]

3 **(a)** The equation of a curve is given by $y = a\sqrt{x} + \frac{b}{x} + cx^2 - d$, where a, b, c and d are constants. It is given that the curve passes through the points $(1,-9)$, $\left(4,\frac{23}{4}\right)$ and $\left(9,\frac{217}{3}\right)$. It is also known that the curve has a stationary point at $(1,-9)$. Determine the equation of the curve. [3]

(b) **(i)** Prove that $4x^2 + 4x + 3$ is positive for all real values of x. [1]

(ii) Solve $\frac{4x^3 + 4x^2 + 3x}{x^2 + x - 2} < 0$. [2]

(iii) Hence, solve $\frac{4\left|\ln x\right|^3 + 4(\ln x)^2 + 3\left|\ln x\right|}{(\ln x)^2 + \left|\ln x\right| - 2} \geq 0$, leaving your answers in exact form. [3]

4 The functions f, g and h are defined for $x \in \mathbb{R}$ by

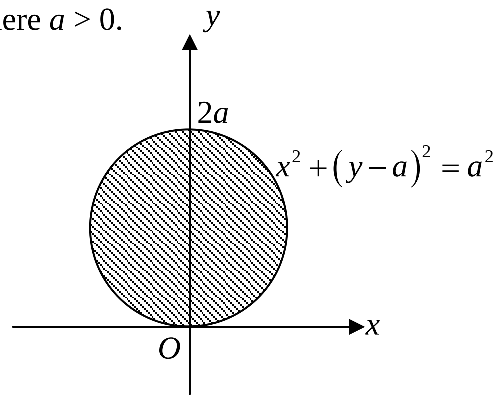
$$\begin{aligned} &\mathrm{f} : x \mapsto x + \frac{1}{x}, \quad x > 0, \\ &\mathrm{g} : x \mapsto -(x+1)^2 + \lambda, \quad -2 < x < 2 \text{ and } \lambda \text{ is a constant,} \\ &\mathrm{h} : x \mapsto \ln(x+2), \quad x > 0. \end{aligned}$$

(i) Give a reason why f does not have an inverse. [1]

(ii) The function f has an inverse if its domain is restricted to $0 < x \leq k$. State the largest value of k and define f^{-1} corresponding to this domain of f. [4]

(iii) Find the minimum value of λ such that the composite function hg exists and find the range of hg in exact form using this value of λ. [4]

5 **(a)** The diagram below shows the region bounded by a circle with equation $x^2+(y-a)^2=a^2$, where $a>0$.



Using integration, show that the volume of revolution formed when the region is rotated through π radians about the y-axis is equal to the volume of a sphere with radius a. [3]

(b) The curve C has parametric equations

$$x=\ln\frac{(t^2-1)^{\frac{1}{2}}}{t}\ ,\ y=t(5t^2-8),\quad \text{where}\quad t>1.$$

Find the exact area bounded by the curve C, the x-axis and the lines $x=\ln\left(\frac{\sqrt{3}}{2}\right)$ and $x=\ln\left(\frac{\sqrt{24}}{5}\right)$. [6]

Section B: Statistics [60 marks]

6 Explain what you understand by the term sample and why it is often desirable to take samples in practical statistical investigations. Give a real-life example of a situation in which random sampling could be used. [4]

7 A machine is set to fill bottles with 1000 ml of soy milk. A random sample of 80 filled bottles was taken and the volume of soy milk in each bottle was measured. The sample mean and standard deviation obtained was 997.7 ml and 11.23 ml respectively. Test, at the 4% level of significance, whether the mean volume of soy milk is 1000 ml. [5]

Explain what is meant by 4% level of significance in the context of the question. [1]

8 **(a)** Find the number of different arrangements of the name "BENNETT TAN"

(i) beginning with B and ending with N, [1]

(ii) with all the T's separated and all the N's separated. [3]

(b) Bennett wants to create an 8-character alpha-numeric password containing 4 letters from his name "BENNETT TAN" and 4 numbers (which can be repeated) from the digits 0-9. How many different passwords can he choose from if

(i) all the letters and numbers are distinct, [2]

(ii) only 2 of the letters are identical and only 2 of the numbers are identical. [2]

9 Cards are drawn at random and without replacement from a deck of 20 cards, which are numbered $1, 2, 3, \ldots, 20$.

(a) Find the probability that of three cards drawn from the deck,

(i) all three cards are even; [1]

(ii) exactly one of the three cards is even. [2]

(b) Let

A be the event that the first card drawn is less than or equal to 5;

B be the event that the second card drawn is greater than or equal to 5.

Find, in any order,

(i) $\mathrm{P}(B)$, [2]

(ii) $\mathrm{P}(A \cap B)$, [2]

(iii) $\mathrm{P}\left(B \middle| A\right)$. [1]

10 Wafflez sells waffles of either cheese or peanut flavour. The number of cheese and peanut waffles sold per 15 minutes may be assumed to be independent Poisson variables with mean 4 and 5 respectively.

(i) Find the probability that at least 10 waffles are sold in a randomly chosen 15-minute period. [2]

(ii) In a randomly chosen 15-minute period, at least 10 waffles are sold. Find the probability that all waffles sold are peanut flavour. [2]

(iii) Find the probability that in a randomly chosen one-hour period, at least 10 waffles are sold in each of the 4 consecutive 15-minute periods. [2]

(iv) Using an approximation, find also the probability that at least 40 waffles are sold in a randomly chosen one-hour period. Explain why you would expect this first probability to be greater than the probability in **(iii)**. [4]

11 **(a)** An experiment was carried out to test the strength of the bond formed by applying a certain type of instant cement to two plastic surfaces. The strength of the bond formed, F Newton, was measured as the time, t seconds, elapsed. The results are given in the table.

F	22	97	260	11	33	19	52	146
t	21	65	86	5	35	12	49	77

(i) Calculate the product moment correlation coefficient between F and t. [1]

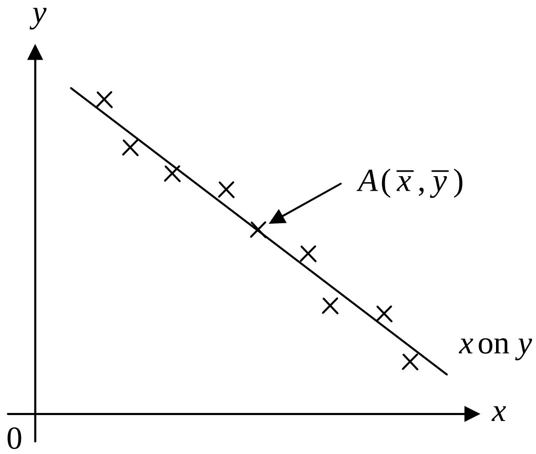
A new variable $x = t^2$ is introduced. For the variables F and x,

(ii) calculate the product moment correlation coefficient, and comment on the suitability of the model in **(i)** in relation to this new value of product moment correlation coefficient, [2]

(iii) calculate the equations of the regression lines of F on x and x on F, [2]

(iv) use an appropriate regression line to estimate the time taken for the instant cement to form a bond with a strength of 100 Newtons. [2]

(b) The diagram below shows the scatter plot and the linear regression line of x on y for a sample of bivariate data. The point $A(\bar{x},\bar{y})$ is marked on the diagram as shown.



It is given that the equation of the linear regression line of x on y is $3x+2y=51$.

(i) A student calculated the equation of the linear regression line of y on x to be $8x-11y=36$. Comment on the validity of the student's answer. [2]

It is further given that the value of the product moment correlation coefficient of the data is -0.943.

(ii) Copy the diagram above on your answer script. Draw on the same diagram, the approximate position of the linear regression line of y on x, indicating clearly the point $A(\bar{x},\bar{y})$ in your diagram. Give reasons to support your answer. [3]

12 **(a)** The height, X cm, of boys in a school may be assumed to follow a normal distribution with mean μ and variance σ^2.
Given that $\mathrm{P}(X<155)=\mathrm{P}(X>185)=0.025$, find the values of μ and σ^2. [4]
Hence, find the probability that

(i) the total height of two randomly chosen boys exceeds twice the height of a third randomly chosen boy by at least 5 cm. [2]

(ii) the mean height of 50 randomly chosen boys is less than 172 cm. [2]

(b) The mass of a randomly chosen box of hamster food has mean μ g and standard deviation 5 g. A large sample of n boxes is taken. Find the least value of n such that $\mathrm{P}\left(\left|\overline{X}-\mu\right|<1\right)>0.99$ where $\overline{X}$ is the sample mean. [4]